ZHENIS DUISSEKOV

Houston, TX • +1 (346) 565-2525 • zduissekov@gmail.com
LINKEDIN • GITHUB • WEBSITE

U.S. Permanent Resident — No Sponsorship Required

PROFESSIONAL SUMMARY

Senior Backend Engineer with 7+ years of experience building distributed systems, event-driven architectures, and high-throughput production services in Go and Python. Experienced in designing scalable APIs, Kafka-based data pipelines, ClickHouse analytics, and modern observability using OpenTelemetry, Jaeger, Prometheus, and Grafana. Proven track record of solving complex production challenges—from reducing processing time from 30 minutes to seconds to productionizing services processing over 1.2M events per day. Currently building end-to-end SaaS products across backend services, integrations, CI/CD, and frontend applications while remaining focused on backend engineering.

CORE SKILLS

Languages: Go, Python, SQL, Node.js

Concurrency: Goroutines, Worker Pools, Context, Synchronization
Messaging & Streaming: Apache Kafka, RabbitMQ, NATS
Databases: PostgreSQL, ClickHouse, Redis, MongoDB
Observability: OpenTelemetry, Jaeger, Prometheus, Grafana, pprof profiling
Infrastructure: Docker, Kubernetes, GitLab CI/CD, Linux
APIs & Architecture: REST, GraphQL, gRPC, Microservices, Event-Driven Architecture, Clean Architecture
AI/LLM: Open/Claude API, local LLMs, AI-assisted development

Testing: Playwright, End-to-End Testing, GitLab CI/CD

PROFESSIONAL EXPERIENCE

Software Engineer | OptiSigns Inc. *Sep 2025 – Present*

Houston, TX • Digital signage SaaS platform used by businesses worldwide

- Developed backend services and GraphQL/REST integrations powering digital signage workflows and enterprise customer deployments.
- Delivered end-to-end product features across Node.js backend services, React/TypeScript frontend, MongoDB, and third-party integrations.
- Modernized legacy Angular applications by migrating them to React/TypeScript, improving maintainability and enabling continued product evolution.
- Owned the end-to-end testing process by maintaining a GitLab CI/CD pipeline executing 900+ Playwright tests across multiple validation stages, while contributing new tests and improving release confidence.
- Developed engineering dashboards to visualize E2E test health, release readiness, and quality trends, enabling faster identification of regressions.
- Diagnosed and resolved complex production issues across backend services, frontend applications, and enterprise customer deployments.
- Partnered with product managers, engineering teams, and customer-facing stakeholders to evaluate technical feasibility and accelerate delivery of new product initiatives.
- Leveraged AI-assisted development tools (Claude, GitHub Copilot) to accelerate debugging, refactoring, and feature delivery.

Backend Engineer | QazCode / Beeline Kazakhstan / VEON Group *Mar 2021 – Aug 2025*

Almaty, Kazakhstan • Telecom-scale internal systems for a NASDAQ-listed operator serving 160M+ customers

- Developed Go-based Kafka consumers using goroutines, worker pools, batching, retry/backoff logic, and context cancellation to process high-volume events reliably under load.

- Productionized a Kafka → ClickHouse pipeline processing 1.2M logs/day — refactored into a stateless, horizontally scalable service with async dual writes, monthly partitioning, and integrity checks via materialized views.
- Re-architected a legacy Go integration service, reducing processing time from 30 minutes to seconds by eliminating cron retries, introducing asynchronous workflows, and simplifying operational recovery.
- Led the team's first OpenTelemetry + Jaeger deployment, exposing root cause latency in mission-critical endpoints and restoring full traceability to production workflows.
- Rebuilt Grafana/ClickHouse observability stack, delivering actionable dashboards (e.g. top 10 failing endpoints) that drove a 6-month initiative to eliminate unstable APIs.
- Prototyped AI-assisted log analysis using OpenAI APIs and local LLMs (Ollama) for anomaly detection and log summarization.
- Raised test coverage from 32% → 46% via SonarQube-tracked refactoring initiative.

Co-Founder & CTO | BAIOL — *Oct 2017 – Feb 2021*

Astana, Kazakhstan • SaaS monitoring startup serving finance, oil & gas, and healthcare clients

- Built and operated a SaaS monitoring platform supporting 100+ servers, 500+ workstations, SNMP-enabled devices, and IoT sensors across multiple enterprise clients.
- Designed and implemented distributed Go and Python services for infrastructure monitoring and automated data collection across enterprise networks.
- Built a COVID reporting system adopted by the city to reduce administrative load on emergency services.
- Delivered 5 real-world pilots; 2 remained in operation long-term. Secured a full-year contract with a local oil institute.
- Led end-to-end delivery: architecture, engineering, client demos, and production support.

EDUCATION

M.Sc. Electrical Engineering — King Abdullah University of Science and Technology (KAUST)

GPA: 3.7/4.0

B.Sc. Electrical Engineering — University of Southern California (USC)

GPA: 3.6/4.0 • BOLASHAK Scholar • Honors

LANGUAGES

English (Technical) • Russian (Native) • Kazakh (Native) • Turkish (Conversational) • Spanish (Elementary)